

Figure 1: Internet of Things (IoT)



Figure 2: Basic requirements of a reference architecture

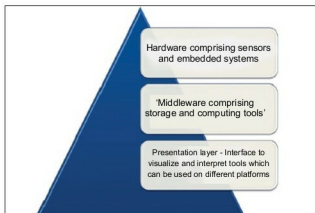


Figure 3: The components of the IoT

The IoT reference architecture

It is difficult to formulate standard architecture for this technology due to the wide variations in devices, network configurations and so on. So let's consider a reference architecture. Requirements for a reference architecture are based on form factors and the power available for the IoT devices. It also depends on how the devices are manufactured and used. The basic requirements for the design of the architecture can be summarised in Figure 2.

IoT architecture comprises three main components as shown in Figure 3.

The basic technologies enabling the IoT are shown in Figure 4.

The IoT uses electromagnetic fields to transfer the data which can be used for tracking and identifying the objects. It basically depends on small electronic devices consisting of a chip and antenna for transmission of data. These chips are capable of transferring huge amounts of data. These devices work just like a bar code scanner. Every object in the network is provided a unique identity. The devices scan the objects in order to identify them, by recognising the unique ID. The advantage of these devices is that they don't need such precise scanning as required when using bar codes.

Now a days, devices are made up of low-power circuits and support wireless communication, which enables them

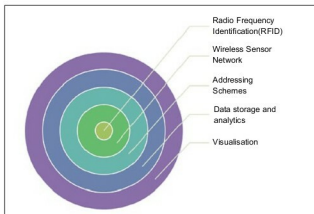


Figure 4: Basic technologies

to be available at a low cost. This enables the creation of devices made up of a large number of intelligent sensors which are used for sensing, data gathering and analysis in various environments. Sensor data is shared between various nodes and then stored at a centralised network for analysis.

The components of a WSN or wireless sensor network are WSN hardware, the communication stack, middleware, secure data aggregation systems, etc.

The success of the IoT depends on each object or device's unique address. Features of this address are uniqueness, persistence, reliability, etc. Handling such a large number of devices requires a platform that is reliable, agile, elastic and easy to use. Such a platform is provided by the cloud. Cloud computing supports the IoT. It is an architecture that provides various technological capabilities such as affordable storage, multi-tenancy, on-demand services, quality of service, a 'pay per use' model and so on. It provides three types of services which are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

IaaS provides a platform to use the hardware, which can be provided to the user as a cloud resource. PaaS provides the platform to access the IoT data and to develop custom applications. And SaaS provides the platform for users to run applications.

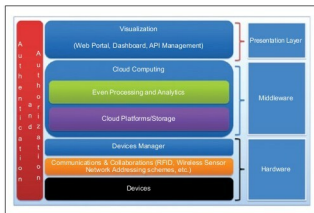


Figure 5: IoT reference architecture